

YULIN LUO

Beijing, China · ✉ yulin@stu.pku.edu.cn · <https://yulin-luo.github.io/>

👤 RESEARCH PROFILE

I am a Ph.D. candidate at Peking University, advised by Prof. Shanghang Zhang. My research focuses on open-world generalization for embodied intelligence, with emphasis on generalizable embodied foundation models, robot-scenario data and benchmark construction, and model-data co-evolution.

🎓 EDUCATION

Peking University, Ph.D. Candidate

Sep. 2023 – Present

- Advisor: Prof. Shanghang Zhang.

Shanghai Jiao Tong University, B.Eng. in Automation

Sep. 2019 – Jun. 2023

- GPA: 3.87/4.30; Major rank: 2/89.

🔬 RESEARCH INTERESTS

- **Model:** building generalizable embodied foundation models for open-world robot intelligence.
- **Data / Benchmark:** constructing embodied-agent benchmarks, multimodal evaluations, and robot-scenario data.
- **Model-Data Co-evolution:** scaling embodied intelligence through data-driven model improvement and model-guided data construction.

📄 SELECTED PUBLICATIONS

1. RoboBench: A Comprehensive Evaluation Benchmark for MLLMs as Embodied Brain
Yulin Luo*, Chun-Kai Fan*, Menghang Dong*, Jiayi Shi*, Mengdi Zhao*, Bo-Wen Zhang*, et al.
First author. *ECCV 2026*
CCF-B
2. LLM as Dataset Analyst: Subpopulation Structure Discovery with Large Language Model
Yulin Luo*, Ruichuan An*, Bocheng Zou, Yiming Tang, Jiaming Liu, and Shanghang Zhang.
First author. *ECCV 2024*
CCF-B
3. Look Before Acting: Enhancing Vision Foundation Representations for Vision-Language-Action Models
Yulin Luo*, Hao Chen*, Zhixuan Wu*, Bo Sui*, Jiaming Liu*, et al.
First author. *arXiv 2026*
4. Decomposing the Neurons: Activation Sparsity via Mixture of Experts for Continual Test-Time Adaptation
Rongyu Zhang*, Aosong Cheng*, **Yulin Luo***, Gaole Dai, Huanrui Yang, et al.
Oral. Co-first author. *AAAI 2026*
CCF-A
5. RoboMIND: Benchmark on Multi-Embodiment Intelligence Normative Data for Robot Manipulation
Kun Wu, Chengkai Hou, Jiaming Liu, et al., **Yulin Luo**, et al. *RSS 2025*
CCF-B
6. RandStainNA: Learning Stain-Agnostic Features from Histology Slides by Bridging Stain Augmentation and Normalization
Yiqing Shen*, **Yulin Luo***, Dinggang Shen, and Jing Ke.
Co-first author. *MICCAI 2022*
CCF-B
7. Efficient Deweather Mixture-of-Experts with Uncertainty-Aware Feature-Wise Linear Modulation
Rongyu Zhang, **Yulin Luo**, Jiaming Liu, Huanrui Yang, et al. *AAAI 2024*
CCF-A

RESEARCH INTERNSHIPS

Simplexity Robotics, Research Intern Dec. 2025 – Present

- Research on embodied intelligence and generalizable robot foundation models.

Beijing Academy of Artificial Intelligence (BAAI), Research Intern Aug. 2024 – Oct. 2025

- Research on embodied-brain benchmarks for multimodal large language models and robot intelligence.

ByteDance, Research Intern Mar. 2024 – Aug. 2024

- Trust & Safety Applied Algorithm Team for Ads and E-commerce.

HONORS AND AWARDS

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|---|------------------|
| • Shanghai Jiao Tong University Outstanding Graduate | Jun. 2023 |
| • Weichai Power Scholarship | Oct. 2022 |
| • Huawei Scholarship | Dec. 2021 |
| • Shanghai Jiao Tong University B-Class Merit Scholarship | 2020, 2021, 2022 |

ACADEMIC SERVICE

- Reviewer for AAAI 2026, ICLR 2026, CVPR 2026, ECCV 2026, and NeurIPS 2026 Datasets & Benchmarks.